

Airline Database SQL Query Suite

Comprehensive 25-Query Reference Manual based on the Entity-Relationship Diagram (ERD)

Basic / Single-Table Queries

1. Retrieve all passengers registered in the database:

```
SELECT * FROM Passenger;
```

2. Find all aircraft with a passenger capacity greater than 200:

```
SELECT Aircraft_id, RegistrationNumber, Model, Capacity  
FROM Aircraft  
WHERE Capacity > 200;
```

3. Get all scheduled or active flights:

```
SELECT FlightNumber, DepartureTime, ArrivalTime, Status  
FROM Flight  
WHERE Status = 'Scheduled';
```

4. List all airports located in a specific country (e.g., 'USA'):

```
SELECT Name, City, IATA_Code  
FROM Airport  
WHERE Country = 'USA';
```

5. Find all bookings made on a specific date:

```
SELECT Booking_Id, Passenger_id, Status  
FROM Booking  
WHERE Bookingdate = '2026-03-15';
```

6. List all pilots who have logged more than 1,000 flight hours:

```
SELECT Employee_Id, LicenseNumber, FlighHours  
FROM Pilot  
WHERE FlighHours > 1000;
```

7. Retrieve all available seats for a specific flight:

```
SELECT SeatNumber
FROM Seat
WHERE Flight_Id = 101 AND Availability = TRUE;
```

8. Find cabin crew members who speak 'Spanish':

```
SELECT Employee_Id, Position, Language
FROM CabinCrew
WHERE Language = 'Spanish';
```

Intermediate / Multi-Table Joins & Aggregations

9. Get all bookings made by passengers, including passenger names:

```
SELECT P.Passenger_id, P.F_Name, P.L_Name, B.Booking_Id, B.Bookingdate, B.Status
FROM Passenger P
JOIN Booking B ON P.Passenger_id = B.Passenger_id;
```

10. Find departure and arrival airport details for every flight:

```
SELECT F.FlightNumber,
       Dep.Name AS Departure_Airport,
       Arr.Name AS Arrival_Airport
FROM Flight F
JOIN Airport Dep ON F.Departs_from_Airport_Id = Dep.Airport_id
JOIN Airport Arr ON F.Arrives_at_Airport_Id = Arr.Airport_id;
```

11. List all flights operated by a specific airline (e.g., 'Air Canada'):

```
SELECT F.FlightNumber, F.DepartureTime, A.Name AS Airline_Name
FROM Flight F
JOIN Aircraft AC ON F.Aircraft_id = AC.Aircraft_id
JOIN AirLine A ON AC.Airline_id = A.Airline_id
WHERE A.Name = 'Air Canada';
```

12. Find tickets and their associated total baggage count:

```
SELECT T.TicketNumber, T.Class, COUNT(Bg.BagNumber) AS Total_Baggage
FROM Ticket T
LEFT JOIN Baggage Bg ON T.Ticket_Id = Bg.Ticket_Id
GROUP BY T.TicketNumber, T.Class;
```

13. Count the total number of flights operated by each aircraft:

```
SELECT Aircraft_id, COUNT(Flight_Id) AS Total_Flights
FROM Flight
GROUP BY Aircraft_id;
```

14. Retrieve pilot details combined with general employee information:

```
SELECT E.Employee_Id, E.Name, E.Phone, P.LicenseNumber, P.FlighHours
FROM Employee E
JOIN Pilot P ON E.Employee_Id = P.Employee_Id;
```

15. Calculate the total number of bookings made per passenger:

```
SELECT P.Passenger_id, P.F_Name, P.L_Name, COUNT(B.Booking_Id) AS Total_Bookings
FROM Passenger P
LEFT JOIN Booking B ON P.Passenger_id = B.Passenger_id
GROUP BY P.Passenger_id, P.F_Name, P.L_Name;
```

16. Retrieve ticket details generated for a specific booking ID:

```
SELECT B.Booking_Id, T.TicketNumber, T.Class, T.IssueDate
FROM Booking B
JOIN Ticket T ON B.Booking_Id = T.Booking_Id
WHERE B.Booking_Id = 501;
```

17. Count total cabin crew staff grouped by position:

```
SELECT Position, COUNT(Employee_Id) AS Total_Staff
FROM CabinCrew
GROUP BY Position;
```

Advanced / Complex Joins, Subqueries & Analytics

18. List complete flight itinerary connecting passenger, flight, aircraft, and airline:

```
SELECT P.F_Name, P.L_Name, F.FlightNumber, AL.Name AS Airline, AC.Model
FROM Passenger P
JOIN Booking B ON P.Passenger_id = B.Passenger_id
JOIN Ticket T ON B.Booking_Id = T.Booking_Id
JOIN Flight F ON F.Flight_Id = B.Flight_Id
JOIN Aircraft AC ON F.Aircraft_id = AC.Aircraft_id
JOIN AirLine AL ON AC.Airline_id = AL.Airline_id;
```

19. Find all personnel (pilots and cabin crew) assigned to work on a specific flight:

```
SELECT E.Employee_Id, E.Name, 'Pilot' AS Role
FROM Employee E
JOIN Pilot P ON E.Employee_Id = P.Employee_Id
JOIN Work_On W ON E.Employee_Id = W.Employee_Id
WHERE W.Flight_Id = 101

UNION

SELECT E.Employee_Id, E.Name, 'Cabin Crew' AS Role
FROM Employee E
JOIN CabinCrew CC ON E.Employee_Id = CC.Employee_Id
JOIN Work_On W ON E.Employee_Id = W.Employee_Id
WHERE W.Flight_Id = 101;
```

20. Find the airline that owns the highest number of aircraft:

```
SELECT AL.Name, COUNT(AC.Aircraft_id) AS Total_Aircrafts
FROM AirLine AL
JOIN Aircraft AC ON AL.Airline_id = AC.Airline_id
GROUP BY AL.Airline_id, AL.Name
ORDER BY Total_Aircrafts DESC
LIMIT 1;
```

21. Identify flights where booked seat count equals total capacity (Fully Booked):

```
SELECT F.Flight_Id, F.FlightNumber, AC.Capacity
FROM Flight F
JOIN Aircraft AC ON F.Aircraft_id = AC.Aircraft_id
JOIN Seat S ON F.Flight_Id = S.Flight_Id
WHERE S.Availability = FALSE
GROUP BY F.Flight_Id, F.FlightNumber, AC.Capacity
HAVING COUNT(S.SeatNumber) = AC.Capacity;
```

22. Find all unique passengers who have booked tickets in 'Business' class:

```
SELECT DISTINCT P.Passenger_id, P.F_Name, P.L_Name, P.Email
FROM Passenger P
JOIN Booking B ON P.Passenger_id = B.Passenger_id
JOIN Ticket T ON B.Booking_Id = T.Booking_Id
WHERE T.Class = 'Business';
```

23. Find airports handling more than 50 departing flights:

```
SELECT AP.Name, AP.IATA_Code, COUNT(F.Flight_Id) AS Total_Departures
FROM Airport AP
JOIN Flight F ON AP.Airport_id = F.Departs_from_Airport_Id
GROUP BY AP.Airport_id, AP.Name, AP.IATA_Code
HAVING COUNT(F.Flight_Id) > 50;
```

24. Find passengers traveling with more than 2 pieces of baggage:

```
SELECT P.Passenger_id, P.F_Name, P.L_Name, COUNT(BG.BagNumber) AS Total_Bags
FROM Passenger P
JOIN Booking B ON P.Passenger_id = B.Passenger_id
JOIN Ticket T ON B.Booking_Id = T.Booking_Id
JOIN Baggage BG ON T.Ticket_Id = BG.Ticket_Id
GROUP BY P.Passenger_id, P.F_Name, P.L_Name
HAVING COUNT(BG.BagNumber) > 2;
```

25. Calculate the average flight duration per airline:

```
SELECT AL.Name AS Airline_Name, AVG(F.Duration) AS Avg_Flight_Duration
FROM Airline AL
JOIN Aircraft AC ON AL.Airline_id = AC.Airline_id
JOIN Flight F ON AC.Aircraft_id = F.Aircraft_id
GROUP BY AL.Airline_id, AL.Name;
```